Steps	Reason
4. $JS(c) \Rightarrow W(c)$	Universal instantiation on 1
5. $JS(c)$	Simplification on 3
6. $W(c)$	Modus ponens on 5,4
7. $CM(c) \wedge JS(c)$	Commutative law on 3
8. $CM(c)$	Simplification on 7
9. $W(c) \wedge CM(c)$	Conjunction on 6,8
10. $\exists x(W(x) \wedge CM(x))$	Existential generalization on 9

d) "There is someone in this class who has been to France. Everyone who goes to France visits the Louvre. Therefore, someone in this class has visited the Louvre."

$\boxed{\text{Soln:}}$ $C(x)$: x belongs to this class $F(x)$: x has visited France

$L(x)$: x has visited the Louvre $\boxed{\text{Domain:}}$ All humans

$\boxed{\text{Premise:}}$ 1. $\exists x(C(x) \wedge F(x))$ 2. $\forall x(F(x) \Rightarrow L(x))$

$\boxed{\text{Conclusion:}}$ $\exists x(C(x) \wedge L(x))$

Steps	Reason
1. $\exists x(C(x) \wedge F(x))$	Premise
2. $\forall x(F(x) \Rightarrow L(x))$	Premise
3. $C(c) \wedge F(c)$ for some c in the domain	Existential instantiation on 1
4. $F(c) \Rightarrow L(c)$	Universal instantiation on 2
5. $C(c)$	Simplification on 3
6. $F(c) \wedge C(c)$	Commutative law on 3
7. $F(c)$	Simplification on 6
8. $L(c)$	Modus ponens on 7,4